

Exercise 2: Aggregate Functions & Operators

SQL Fundamentals

Table 1: students

01 The students Table

A list of students enrolled at the academy, with their age and department.

TABLE students

student_id	name	age	department
1	Alice	20	IT
2	Bob	22	HR
3	Charlie	21	IT
4	Diana	23	Finance
5	Eve	22	HR

Question 1

01 List all distinct departments in the students table.

EXPECTED COLUMNS: department

SQL code:

Run all (1000) ✓ Just now (13s) workspace default

```
4 --Question 1: List all distinct departments in the students table; EXPECTED COLUMNS:
  department
5 SELECT DISTINCT department
6 FROM brightlearn.master_data.students;
7
```

Add parameter

Table	
1	IT
2	HR
3	Finance

Question 2

02 Get the average age of students per department.

EXPECTED COLUMNS: **department** **avg_age**

SQL code:

▶ Run all (1000) ✓ Just now (1s) workspace default ▼

```
8  --Question 2: Get the average age of students per department, EXPECTED COLUMNS:
   department avg_age
9  SELECT department,
10         AVG(age) AS avg_age
11  FROM brightlearn.master_data.students
12  GROUP BY department;
```

Add parameter

Table ▼ +

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	department	avg_age
1	IT	20.5
2	HR	22
3	Finance	23

Question 3

03 Show departments with more than 1 student.

EXPECTED COLUMNS: **department** **student_count**

SQL code:

▶ Run all (1000) ✓ Just now (1s) workspace default ▼

```
14 --Question 3: Show departments with more than 1 student, EXPECTED COLUMNS: department
   student_count
15 SELECT department,
16        COUNT(*) AS number_of_students
17  FROM brightlearn.master_data.students
18  GROUP BY department
19  HAVING COUNT(*) > 1;
```

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	department	number_of_students
1	IT	2
2	HR	2

Question 4

04 Get all students whose age is between 21 and 23.

EXPECTED COLUMNS: student_id name age department

SQL code:

▶ Run all (1000) ✓ Just now (1s) workspace default

```
21 --Question 4: Get all students whose age is between 21 and 23, EXPECTED COLUMNS:
22   student_id name, age, department
23 SELECT name,
24         age,
25         department,
26         student_id
27 FROM brightlearn.master_data.students
28 WHERE age BETWEEN 21 AND 23;
```

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	A _C name	I ₃ age	A _C department	I ₃ student_id
1	Bob	22	HR	2
2	Charlie	21	IT	3
3	Diana	23	Finance	4
4	Eve	22	HR	5

Question 5

05 List all students in the IT or HR department who are older than 21.

EXPECTED COLUMNS: student_id name age department

SQL code:

▶ Run all (1000) ✓ Just now (1s) workspace default

```
28
29 --Question 5: List all students in the IT or HR department who are older than 21,
30   EXPECTED COLUMNS: student_id, name, age, department
31 SELECT name,
32         age,
33         department,
34         student_id
35 FROM brightlearn.master_data.students
36 WHERE department IN ('IT', 'HR')
37        AND age>21;
```

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	A _C name	I ₃ age	A _C department	I ₃ student_id
1	Bob	22	HR	2
2	Eve	22	HR	5

Table 2: courses

02 The courses Table

A catalogue of the courses offered by each department, along with the credits awarded.

TABLE courses			
course_id	course_name	department	credits
101	SQL Basics	IT	3
102	Python	IT	4
103	Data Science	IT	4
104	Excel	Finance	2
105	Statistics	HR	3

Question 6

06 Show total credits per department, only for departments with more than 5 total credits.

EXPECTED COLUMNS: department total_credits

SQL code:

Run all (1000) ✓ Just now (<1s) workspace default Edit

```
37
38 --Question 6: Show total credits per department, only for departments with more than 5
39 --total credits, EXPECTED COLUMNS: department, total_credits
40 SELECT department,
41      SUM(credits) AS total_credits
42 FROM brightlearn.master_data.courses
43 GROUP BY department
44 HAVING total_credits>5;
45
```

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	department	total_credits
1	IT	11

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Question 7

07 List all courses that do not have 4 credits.

EXPECTED COLUMNS: course_id course_name department credits

SQL code:

▶ Run all (1000) ✓ Just now (1s) workspace default Edit

```
44
45  --Question 7: List all courses that do not have 4 credits, EXPECTED COLUMNS: course_id,
46  course_name, department, credits
47  SELECT course_id,
48          course_name,
49          department,
50          credits
51  FROM brightlearn.master_data.courses
52  WHERE credits <> 4;
```

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	1 2 3 course_id	A B C course_name	A B C department	1 2 3 credits
1	101	SQL Basics	IT	3
2	104	Excel	Finance	2
3	105	Statistics	HR	3

Question 8

08 Show the top 3 courses by credits in descending order.

EXPECTED COLUMNS: course_id course_name credits

SQL code:

▶ Run all (1000) ✓ Just now (1s) workspace default Edit ⓘ

```
52
53  --Question 8: Show the top 3 courses by credits in descending order, EXPECTED COLUMNS:
54  course_id course_name credits
55  SELECT course_id,
56          course_name
57  FROM brightlearn.master_data.courses
58  ORDER BY credits DESC
59  LIMIT 3;
```

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	1 2 3 course_id	A B C course_name
1	102	Python
2	103	Data Science
3	101	SQL Basics

Table 3: enrollments

03 The enrollments Table

A record of which students are enrolled in which courses, with the grade each student received.

TABLE enrollments			
enrollment_id	student_id	course_id	grade
1	1	101	85
2	2	102	78
3	3	103	90
4	4	104	88
5	5	105	82

Question 9

09 Get the maximum, minimum, and average grade across all enrollments.

EXPECTED COLUMNS: max_grade min_grade avg_grade

SQL code:

▶ Run all (1000)
✓ Just now (1s)
workspace
default

```

59
60 --Question 9: Get the maximum, minimum, and average grade across all enrollments,
61   EXPECTED COLUMNS: max_grade min_grade avg_grade
62   SELECT MAX(grade) AS max_grade,
63          MIN(grade) AS min_grade,
64          AVG(grade) AS avg_grade
65   FROM brightlearn.master_data.enrollments;

```

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Table	1.2 max_grade	1.2 min_grade	1.2 avg_grade
1	90	78	84.6

Question 10

10 Count how many enrollments exist per course.

EXPECTED COLUMNS: course_id enrollment_count

SQL code:

▶ Run all (1000) ✓ Just now (<1s) workspace default Edit

66 --Question 10: Count how many enrollments exist per course, EXPECTED COLUMNS: course_id
enrollment_count
67 SELECT course_id,
68 COUNT(*) AS enrollment_count
69 FROM brightlearn.master_data.enrollments
70 GROUP BY course_id;

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	1 3 course_id	1 3 enrollment_count
1	101	1
2	102	1
3	103	1
4	104	1
5	105	1